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Diagnosis of Breast Cancer using Machine Learning Techniques -A Survey

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Abstract

Breast cancer is a disease in which the cells of the breast develop unnaturally and uncontrollably, resulting in a mass called a tumor. If lumps in the breast are not addressed, they can spread to other regions of the body, including the bones, liver, and lungs. Men and women are both affected by breast cancer, albeit men are at a lower risk. This research investigates the detection of breast cancer by applying machine learning algorithms, deep learning algorithms, and hybrid machine learning approaches to a variety of datasets. These datasets include breast cancer databases from Wisconsin as well as mammography imaging datasets. The goal of this research is to find the best model for breast cancer diagnosis.

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Keywords: Type your keywords here, separated by semicolons ;

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1. Introduction

Breast cancer was diagnosed in 7.8 million women in the preceding three years, according to the World Health Organization (WHO), making it the most frequent cancer worldwide by the end of 2020. Breast cancer causes uncontrolled growth of cells in the breast. Globally, 4–5.5 percent of new cancer cases are recorded each year, increasing morbidity [1]. It has been proven that early detection and treatment of cancer can improve the patient's chances of survival. Breast cancer is detected via mammography. Mammograms are created by a radiologist using

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equipment. The three main components of the breast are lobules, connective tissue, and ducts. Breast cancer usually begins in the ducts or lobules. A breast lump or thickening, changes in size, shape, dimpling, redness, pitting, change in nipple appearance, and abnormal nipple discharge are all signs of breast cancer. A cancer tumor is a type of cell that grows improperly and invades the body's surrounding tissues. Breast cancer tumors can be categorized into two types, namely benign and malignant. For the detection and prediction of breast cancer, a variety of ML algorithms, EML algorithms, and deep learning techniques are available. Deep learning and ML algorithms can be used to predict and find breast cancer diagnoses. These algorithms use the performance of each classifier to find the best outcome. The remainder of the paper is structured as follows: Section 2 presents a dimension reduction approach in preprocessing. Section 3 explains machine learning techniques for breast cancer diagnosis. Section 4 explains how the method's performance is measured. The paper's conclusion is in section 5.

2. Dimension Reduction

A dimension is defined as the number of features in the dataset, and reducing the dimensions in the dataset is called a dimension reduction technique. Feature selection and feature extraction are the two methods used for dimension reduction.

2.1. Feature Selection Technique

Feature selection is a strategy for the extraction of the most significant and helpful features from a dataset to improve the model's accuracy. Three phases are necessary for feature selection: the first is filtering, the second is wrapping, and the third is embedding. [17] classified breast cancer diagnoses on Wisconsin (Original) Breast Cancer (WBC) [52] and Wisconsin Diagnosis of Breast Cancer (WDBC) [54] using feature selection and genetic programming (GP) techniques and achieved accuracy of 100% and 98.24 %, respectively.

2.2. Feature Extraction Technique

Feature extraction technique is used to transform the raw data to a suitable form because the original raw data cannot be applied by the ML model. Principal component analysis (PCA) was employed to extract features in [16] which achieved an accuracy of 99.01 % with the help of MLP model and [27] found 97.52 % result by RF classifier. PCA is a technique for identifying the patterns and correlations in any dataset so that it may be transformed into a dataset without losing any information. PCA and auto encoder are used to extract features in [2]. The auto encoder is a form of neural network that has three components namely an encoder, a code, and a decoder. The encoder is used to compress the input and generate code, then it is used to reconstruct the input.

The multilayer perceptron (MLP) was used to extract features in [34]. An MLP is made up of three layers namely input, hidden, and output layer. MLP has a complete neural network connection. Feature selection is a technique for selecting only important and useful features from a dataset to increase the accuracy of the model. For feature selection, three steps are required. The first one is filter, the second is wrapping, and the last one is embedding.

3. Algorithms and Techniques

3.1. Machine Learning (ML) Classifier

There are numerous machine learning algorithms available, but it is impossible to determine which one is superior to the others because ML techniques are dependent on the data set provided. Researchers utilize the following ML classifiers.

3.1.1. Logistic Regression(LR)

LR is a supervised ML technique. It is used to solve the classification problem (binary or multi-class). LR is based on the concept of probability. The graph representation of LR is:

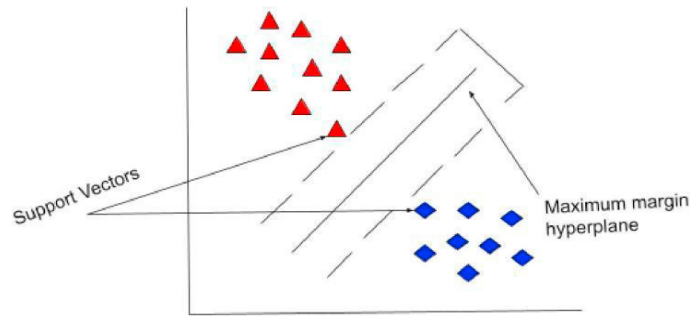


Fig 1. Support Vector Machine.

3.1.2. Support Vector Machine (SVM)

SVM is a supervised ML algorithm that uses the concept of classification as well as regression problems. With the help of hyperplanes, SVM clearly sorts data points into N different groups. The graph representation of SVM is depicted in fig 1. In [4] and [8] SVM classifier achieved the highest accuracy compared to other ML classifiers and simultaneously got 96.5 % and 96.25 accuracy. In [49], a hybrid, SVM, and grey wolf optimization (GWO) were applied to two different datasets: the WDBC dataset, which achieved 98.60 % accuracy, and the Electronic Health Record (EHR) dataset, which achieved 93.26 % accuracy. GWO is a population-based meta heuristic algorithm. A meta heuristic is a standard, independent algorithm framework that finds the best solution out of all possible solutions to an optimization problem.

3.1.3. Decision Tree (DT)

DT is used in the concept of classification as well as regression. DT uses the tree structure, which contains two nodes, namely the decision node and the leaf node. The decision node is nothing but a test, and the classification happens in the leaf node. The tree representation of the decision tree is given in figure 2. In [13], two DT classifiers, J48 and a simple CART algorithm, were used to achieve a 98.13 % accuracy. J48 is based on Iterative Dichotomiser 3 (ID3) and is used to examine the data categorically and continuously. CART is based on Gini's in-purity index.

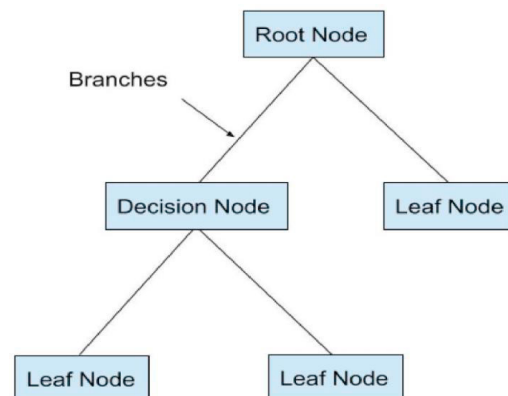


Fig. 2. Decision Tree.

3.1.4. Naive Bayes (NB)

It is based on conditional probability, which means finding the probability of any given variable X happening, given that Y has occurred. In [21], three datasets were used from the UCI ML repository, and for diagnosis, several techniques were applied and also compared with each other.

3.1.5. K Nearest Neighbor (KNN)

KNN is a supervised learning classifier that makes predictions about the grouping of a single data point based on proximity. It can be used for both regression and classification issues, but it is most widely utilized as a classification technique, based on the idea that comparable points can be discovered close together. In [4], a hybrid technique called KNN and Genetic Algorithm (GA) were used, which performed better than other techniques. GA is generally helpful for finding the best solution to any particular optimal problem. In [12] and [32] several ML techniques were compared with each other on the Wisconsin dataset [52] and it was found that the SVM and MLP achieved the highest accuracy simultaneously at 98.24 % and 99.04 %.

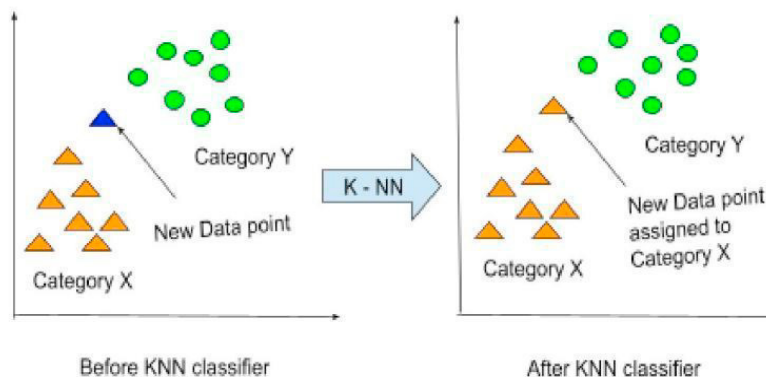


Fig. 3. K Nearest Neighbour.

3.2. Ensemble ML Classifier (EML)

EML is a machine learning technique that predicts the output based on a combination of several ML models. It also improves the performance of a unique model by combining individual models together. The ensemble uses two types of methods: one is **bagging**, and the second is **boosting**. In [23] Random Forest (RF) classifier was applied to the Mammographic Image Analysis Society (MIAS) dataset [55] for breast cancer diagnosis and achieved an accuracy of 97.23 %. **RF is a bagging EML classifier** which randomly selects records from the collected dataset, then creates different datasets and puts all records in these datasets with replacement. With the help of majority voting, the final outcome is predicted. In [10] ML classifier SVM, RF, LR, KNN and stacking classifier were used for breast cancer diagnosis. Stacking classifier combine SVM and RF classifier via LR meta classifier and achieved accuracy of 97.20%. In [19] all ML classifiers and XGBoost classifiers were used for breast cancer diagnosis, XGBoost classifier produced better results and gave the highest accuracy of 98.24 %. For tabular data, the XGBoost classifier is utilized. It's a gradient boosting approach that uses distributed computation to give optimization. [27] used PCA for feature extraction and for classification two techniques called DT and RF were applied on WDBC and found that the RF outperforms the DT technique and achieved 97.52 % accuracy. In [50] individual ML classifier and majority voting ensemble technique were used and found that the ensemble technique based on SVM, KNN and LR gives highest accuracy of 98.1 % with error rate of 0.01 % than individual ML classifier. In this study, we can see that ensemble techniques perform better than individual machine learning techniques.

3.3. Neural Network (NN)

NN is built using three separate layers with a feed-forward design. This is the most commonly used network design today. This network's input layer consists of a collection of input units that accept the elements of input feature vectors. With the hidden units, the input units (neurons) are entirely connected to the hidden layer. The output layer is also fully coupled to the hidden units (neurons). The neural network's response to the activation pattern applied to the input layer is provided by the output layer. The data fed into a neural network is passed layer by layer from the input layer to the output layer via one or more hidden layers. The flow diagram of NN is shown in

Fig 4. In [7] Deep NN are used for reducing the overfitting problem and achieved the F1 score of 98. In [15] hybrid technique of unsupervised ANN, which is also called self-organizing map (SMO) and supervised classifier stochastic gradient descent (SGD), was applied to the WDBC dataset and achieved an accuracy of 99.68 %. [16] used a computer-aided system (CAD) based on joint variable system and constructive DNN which applied to the WDBC dataset and SEER 2017 dataset and gets simultaneously 99.1 % and 89.3 % accuracy. In [21] ANN techniques were compared with ML techniques and it was found that ANN gave the highest accuracy of 99.73 %.

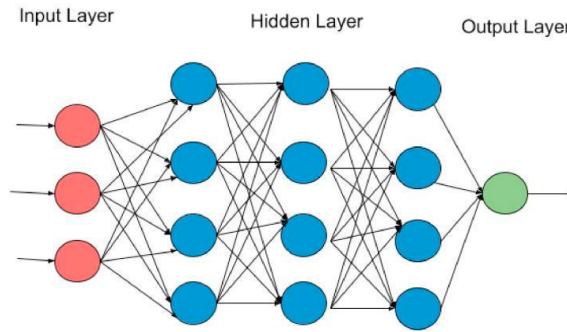


Fig. 4. Neural Network.

3.4. Convolutional Neural Network (CNN)

CNNs are similar to classic ANNs in that they are made up of neurons that learn to optimise themselves. Countless ANNs will still receive input and perform an operation for each neuron. The network will still express a single perceptual scoring function from the input raw picture vectors to the final output of the class score. The last layer will have loss functions connected with the classes, and all of the standard ANN tips and tactics will still apply. The working process of CNN is given in figure 4. In CNN, less preprocessing is required as compared to other ML classifiers. In [6] found 99.67 % result, [7] found F1 score as 98 and [18] achieved 98 % accuracy CNN was applied to the WDBC dataset, but [18] was also compared with the ML classifier. In [36] CNN model was applied on a 2Dmatrix and, for breast cancer diagnosis, the WDBC dataset was used. In [22], CNN model was achieved an accuracy of 98.6 % on BreKHis (Breast Cancer Histopathological Dataset), but in [26] A handcraft feature extraction technique and CNN-SVM technique were developed for detection of breast cancer on BreKHisdataset . In [24] CNN model was used, which was already pre-trained applied on the MIAS dataset, and we observed that VGG16 outperforms the other CNN models and achieved the highest accuracy of 98.96 %.

[25] Developed DCNN models for feature extraction and a Dual Network orthogonal low ranking learning (DOLL) technique for feature selection on hematoxylin and eosin stained breast histology microscopy image dataset from ICIAR 2018 dataset [56] which found an accuracy of 97.70 %. Then, to classify the breast cancer, an ensemble SVM classifier was used, but in [30] and [35] two approaches gradient boosting and pooling operations were used for classification after feature extraction and achieved an accuracy of 93.8 % and 92.50 %. With the usage of weights, the transfer learning technique was employed with a CNN model (InceptionV3) to get the maximum accuracy in [39].

In [29] DL approach was applied to two datasets. The first one is the Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM) [58] and the second one is the INBreast dataset of full field digital mammography (FFDM) images. In [41], the FFDM dataset was used for classification with the help of a shallow DNN model and found an accuracy of 90 %. In [40] U - Net structure based DNN architecture were used which giving the accuracy of 94.31 % and 95.01 % for respectively microcalcification and masses. U-Net structure is a segmentation technique of biomedical images in CNN. In [38] 3D CNN and multi scale curriculum

learning strategy were used to classify the breast cancer diagnosis on dataset of Magnetic resonance imaging (MRI) and also Retina U-net, mask R - CNN as well as radiologist models are compared.

In [42] ensemble CNN model was applied to four different datasets. The first one is hematoxylin and eosin stained breast histology microscopy image dataset from the ICIAR 2018 dataset that achieved an accuracy of 95 %, second one is the BreasKHis dataset [57] and achieved an accuracy of 98.13 % accuracy, third one is the patch Camelyon dataset that achieved an accuracy of 94.64 % and the last one is Bio imaging dataset that achieved an accuracy of 83.10 %.

In [46] a clinic decision support system called Man and Machine Mammography (MAMO), which contains twocomponents, the first one is Multi View CNN, and the second one is Multitask Learning (MTL), was applied to the Tommy dataset that was collected through six NHS Breast Screening Program (NHSBSP) centers all around the United Kingdom [60].

[48] Used CNN hyper-parameters fine-tuning optimization algorithm with help of a tree parzen estimator on thermal image and for breast cancer diagnosis DMR - IR dataset used and achieved an accuracy of 92 %. This model was also compared with pretrained CNN models.

In [51] Y - Net architecture was used for better segmentation and for improvement of accuracy, also compared with ML Techniques. Y-Net model is an advanced implementation of U-Net architecture. for check the performance of model Bio-spy images dataset used and achieved an accuracy of 70 %.

4. Performance metrics

4.1. Accuracy

The number of correct predictions made as a percentage of all predictions made is called classification accuracy. It's only useful when each class has an equal amount of data and all forecasts and prediction errors are equally important, which isn't always the case.

$$Accuracy = \frac{TP+TN}{(TP+TN+FN+FP)} \quad (1)$$

Table 1. Results are compared of Authors' work in the field of Diagnosis of Breast Cancer.

Ref.	Dataset	Approach	Results	Pros	Cons	Year
[2].	WBC, WDBC	SVM, PCA, Autoencoder	99.91 %	Separate the weak features suitable for WBC data set	DL can be applied to improvement	2021
[3].	WDBC	EML algorithm	98.50%	Implementation costs are reduced	Only a limited number of features are employed	2018
[4].	WDBC	ML classifier	SVM and RF accuracy96.50%	A number of ML classifiers are compared	This model was tested on a tiny collection of data	2021
[5].	WDBC	Hybrid model	99%	High accuracy when compare to individual models	Model with a high level of complexity	2017
[6].	WDBC	CNN	99.67%	Image data and big data sets can also be used	A smaller amount of features are employed	2021
[7].	WDBC	DL, NN.	F1 score is 98	Reduce the over-fitting problem	Less number of references used	2021
[8].	WDBC	ML techniques.	SVM accuracy is 96.25%	There is less complication	Be late in determining the outcome	2021
[9].	WDBC	ELM and RBF	99.23%	Reduce the number of errors and improve accuracy	The robustness of deep learning were compared	2021
[10].	WDBC	Stacking classifier	97.20%	-	High level of complexity	2021

[11].	WDBC	EML	99.24%	Here the only thing that is necessary is efficiency, not a categorization parameter	The outcome of certain people were predicted erroneously	2021
[12].	WDBC	LR, SVM, KNN, NV	98.24%	Minimize the cost and time it takes to diagnose	No literature study is available to describe	2020
[13].	WDBC	J48, simple CART, NB, Bayesian LR	98.13%	-	Take time for diagnosis	2018
[14].	WDBC	ANN, DOM	99.686%	-	High complexity	2021
[15].	WDBC	NN	96.00%	Calculation is less expensive and takes less time	Obtaining the dataset for training and testing is difficult	2021
[16].	WDBC, SEER 2017	PCA, MLP, CNN	99.10% and 89.30%	-	Get lower value in confusion matrix	2019
[17].	WDBC	ML technique with genetic programming	Accuracy 100% for WBC, 98.24% for WDBC	In the form of a table each algorithm and measure is thoroughly explained	-	2021
[18].	WDBC	CNN	98%	There is little comparison	There is a lot of complication	2021
[19].	WDBC	XGBOOST	98.24%	There is intricacy but there is a lot more precision	It isn't appropriate for all stages of breast cancer	2021
[20].	WDBC	ANN	99.73%	Applied to a data set of photos as well	-	2021
[21].	WDBC, WBC, WPBC	J48, MLP, NV, SMO, IBK	97.71%	Each data set was subjected to both and ensemble and an individual classifier	For prediction, more than two classifier ensembles are used, which increases complexity	2012
[22].	BreaKHis data set	CNN	98%	The accuracy of this method is superior than that of other machine learning algorithms	A small amount of features are employed	2018
[23].	MIAS data set	RF	97.32%	Increase the rate of cancer identification in areas where cancer masses are hidden	There are less comparisons	2021

4.2. Recall or sensitivity

The recall is determined by dividing the total number of positive samples by the number of positive samples accurately categorized as positive. The higher the recall, the greater the number of positive samples found.

$$Recall = \frac{TP}{(TP+FN)} \quad (2)$$

4.3. Specificity

The number of negatives returned by the ML model can be defined as this. A confusion matrix with the following formula can be used to determine this:

$$Specificity = \frac{TN}{(TP+FN)} \quad (3)$$

4.4. Precision

Precision is calculated as the ratio of correctly predicted positive samples to the total number of positive samples. The precision of the model in categorizing a sample as positive is measured.

$$\text{Precision} = \frac{TP}{(TP+FP)} \quad (4)$$

4.5. F1 Score

The F1 score is a classification error metric that, like any other error metric, aids in the evaluation of algorithm performance. It allows us to assess the machine learning model's performance in terms of binary classification.

$$\text{F1 Score} = \frac{2 * (\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (5)$$

5. Conclusion

The performance of ML algorithms such as SVM, KNN, RF, DT, NB, LR, ELM, and DL algorithms used by various academics is evaluated and compared in this paper. The accuracy of any method is determined by the total number of features and methodology used. Features are utilized as the prediction input. It has also been observed that a dataset is required in the application of ML algorithms for improved accuracy. However, one of the primary obstacles to using ML algorithms is the availability of sufficient data. Different feature extraction methods such as PCA, autoencoders, and MLP were applied to pick attributes, clean data, transform data, and remove missing data from datasets. It is also stated that, while ANNs can be utilized for prediction, they do not produce accurate results because of limitations in the number of layers and neurons. Among all classifiers, the SVM classifier has the highest accuracy for breast cancer categorization. In future, dimension reduction techniques can be used for reducing the dimensions in dataset and Ensemble ML techniques can be used for improving the performance of individual model.

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